Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [MSCCH102]

FIRST SEMESTER M.Sc. (CHEMISTRY) EXAMINATION, DECEMBER 2015

ORGANIC CHEMISTRY

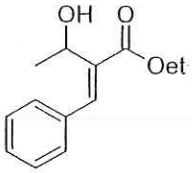
Time : 03 Hours **Maximum Marks:80**

Instructions:
1. Answer **all** Questions
2. Questions **1 to 4** carry **15** marks each.
3. Question No. **5** carries **20** marks.

1a. Outline the conformational analysis of 1,2, 1,3 and 1,4 substituted cyclohexane derivatives.

1b. Write a detailed note on geometrical isomerism in cyclic systems with suitable examples.

1c. Assign E & Z configuration to the following alkenes. **6+6+3=15**



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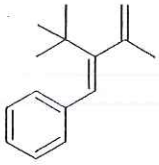

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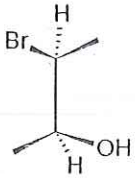
OR

1d. Explain in detail the elements: plane, centre and axis of the symmetry with suitable examples and the conditions necessary for optical isomerism.

1e. How are enantioselective reactions different from enantiospecific reactions? Explain with suitable examples.

1f. Assign the R & S configuration for the below structures. **7+4+4=15**



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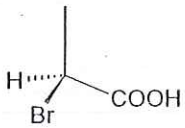

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2a. Explain any four types of reaction mechanisms with suitable examples.

2b. What are pH and pKa values? Explain the relation between them.

2c. What is a molecular fission reaction? Explain the different types involved. **8+4+3**

OR

2d. Outline the structure, formation and stability of free radicals.

2e. How are arynes generated? Give an account of the structure, stability and reactions of arynes.

2f. What are carbanions and carbocations? Explain with examples. **6+5+4=15**

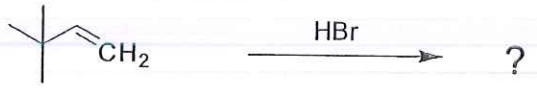
3a. Differentiate between an aromatic and an antiaromatic compound. Explain with an example.

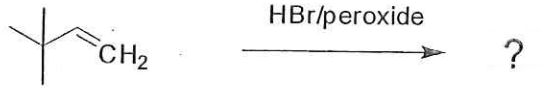
3b. What is activation energy? Explain the thermodynamic effect on activation energy.

3c. Explain the MOT of π electrons of cyclopenta-di-ene carbocation and cyclopenta-di-ene carbanion and explain their stability. **4+5+6=15**

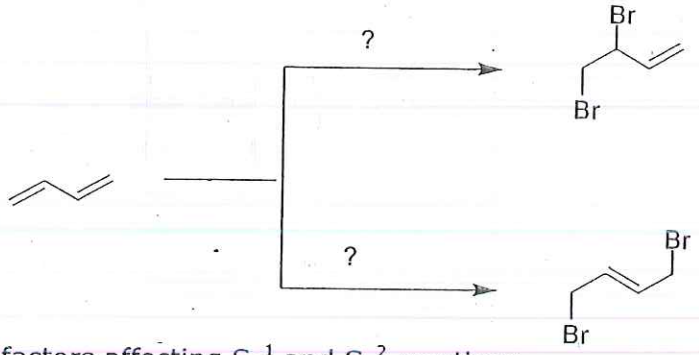
OR

3d. Write the expected products in the below mentioned reactions. Explain the mechanism involved.





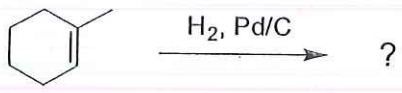
3e. Write the suitable conditions to get 1,4- and 1,2-dibromo compound in the bromination of 1,3-butadiene. Show the mechanism of its formation.



3f. Discuss the important factors affecting S_N^1 and S_N^2 reactions. **5+5+5=15**

4a. Differentiate between a singlet and a triplet state carbene.

4b. Write the product formed in this reaction with its absolute configuration. Is the product formed optically active?

a. 

4c. Pyridene has an even pair of electrons and is still considered an aromatic. Why?

4d. Benzoic acid is stronger than phenol. Give reasons.

4e. Give any two methods of formation/generation of carbocations.

4f. Differentiate an enantiomer from a diasterioisomer with examples.

4g. What is Vilsmeier-Haack reaction? Explain its reaction mechanism. **2x6+3=15**

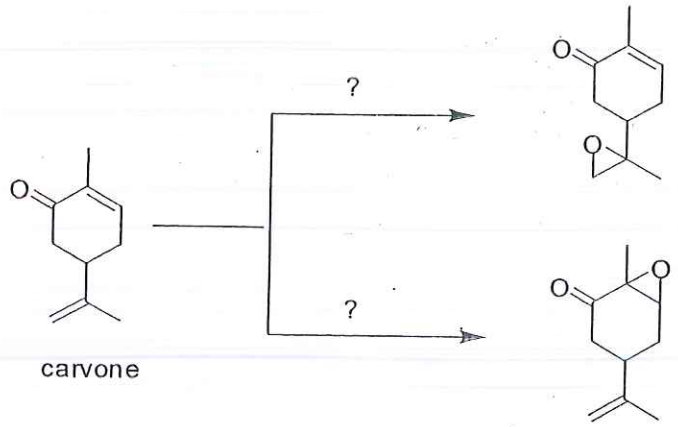
OR

4h. What are non-classical carbocations?

4i. What are nitrenes? Explain with examples.

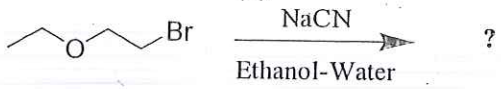
4j. What is optical isomerism?

4k. Identify the required reagents for carrying out the selective epoxidation of carvone.



4l. For each of the chemical substitution reactions below identify the major product and the type of SN reaction (SN^1 or SN^2).

(a)



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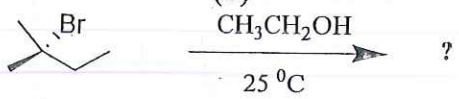


4m. What is aromaticity?

4n. What is a rearrangement reaction? Explain with an example. **2x6+3=15**

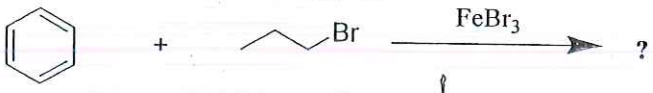
5. **Answer any FOUR of the following.** **4x5=20**

a. Explain the Canh-Ingold-prelog sequence rule involved in designation of configuration to chiral compounds with suitable examples.

b. Explain cycloaddition reactions involving carbene with stereochemistry aspect.

c. Predict the product for the following reactions.

(a)



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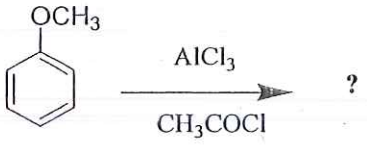

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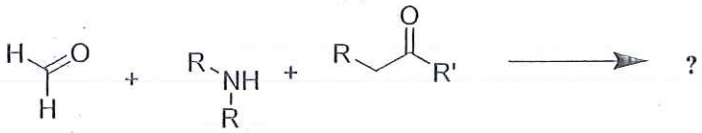
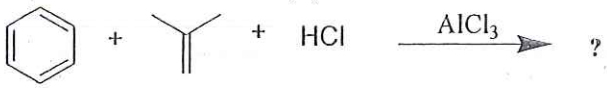


(d)



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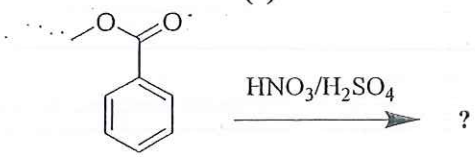
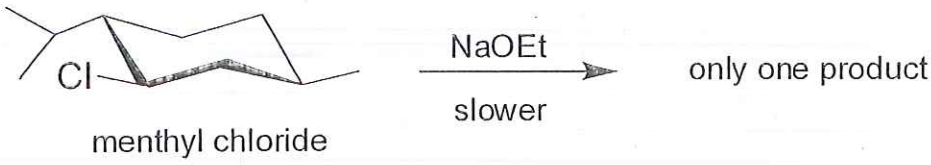
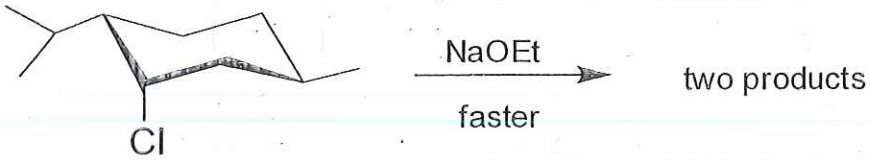


Methyl Benzoate

d. The reaction of menthyl chloride with NaOEt is slow and gives only one product whereas the same reaction of neomenthyl chloride is faster and produces two products (3:1). Explain why? Write the structure of the products obtained from both the isomers.



menthyl chloride



neomenthyl chloride

e. Aromatic compounds are more stable as compared to anti aromatic compounds. Explain this with molecular orbital theory.

f. Explain the inductive effect on strength of organic acids and bases.